



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Instrumentation Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Third Year

Course: Automatic Control System

Course Code: PCC-IN303

Date: 14.10.2025

Time: 11:00 AM to 11:45 AM

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. **Question 1 compulsory and solve any three questions from Q.2, 3, 4 and 5.**

Q.N.		Marks	CO	BT
Q.1	Explain the type of system and derive expressions of steady state errors with step, ramp and parabolic inputs for type 0, type 1 and type 2 systems respectively.	8	CO2	BT2
Q.2	Consider the following under-damped second order system having transfer function given as, $T.F. = \frac{1}{16s^2 + s + 1}$ Find out Delay time (t_d), Rise time(t_r), Peak time(t_p) and Maximum overshoot (M_p)	4	CO2	BT1
Q.3	Step response of system given as i) e^{-2t} ii) $1 - e^{-2t}$ Find out impulse response	4	CO2	BT1
Q.4	For the system with the following Transfer Functions find the initial and final values of the response to a step input $\text{i) } T.F. = \frac{10}{s^2 + 4}$ $\text{ii) } T.F. = \frac{10(s-2)}{s^2 + 4s + 5}$	4	CO2	BT1
Q.5	Find out steady state errors of the following systems with given inputs i) $2u(t)$ and ii) $3tu(t)$ $G1(s) = \frac{10}{s+2}$ $G2(s) = \frac{10}{s(s+2)}$	4	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Develop mathematical models of physical systems using differential equations and Laplace transforms.
CO 2	Analyze the transient and steady-state response of linear time-invariant systems.
CO 3	Apply stability criteria (Routh-Hurwitz, Nyquist, Root Locus, and Bode) to evaluate system stability.
CO 4	Design controllers using classical methods (PID, root locus, frequency response).
CO 5	Implement control system designs using simulation tools like MATLAB and Simulink.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Instrumentation Engineering In-Semester Examination II Academic Year: 2025-26 (Semester – I)	
	Class: TY B.Tech	Subject: Digital Signal Processing Subject Code: PCC-IN 303
Date: 17.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
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- Exchange/sharing of stationery, calculator, etc. is not allowed
- Non Programmable calculator is allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	Find discrete time fourier transform of the given signal $x(n) = \{ \underset{\uparrow}{1}, -1, 2, 2 \}$	4	CO2	BT1
Q.2	A causal discrete time LTI system is described by $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$ Determine a) System Function b) Impulse Response	6	CO2	BT5
Q.3	Find IDFT of given sequence using IDFT formula. $X(K) = \{ 6, 0, -2, 0 \}$	4	CO2	BT1
Q.4	Compute the circular convolution using DFT and IDFT method for following sequences. $X1(n) = \{ 2, 1, 2, -1 \}$ and $X2(n) = \{ 1, 2, 3, 4 \}$	6	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To understand benefits and limitations of processing signals digitally and properties of discrete-time LTI systems.
CO2	Represent and analyze the discrete-time signals and LTI systems in the frequency domain using Discrete-Time Fourier Transform (DTFT), z-transform and Discrete Fourier transform (DFT) tools.
CO3	To Implement DFT efficiently using Fast Fourier Transform (FFT) algorithms and use in practical applications.
CO4	Design an FIR or IIR filter for the specifications given in frequency domain
CO5	To realize digital system whose coefficients are known using hardware or software.
CO6	To propose and design a digital system for simple real application.